

Ai Driven Diet Plainning Chatbot

A Minor Project Synopsis Submitted To



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Submitted By

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Abstract

Unhealthy eating habits, lack of nutritional awareness, and the growing prevalence of lifestyle diseases have increased the demand for accessible diet guidance. Many individuals struggle to identify suitable meals due to limited knowledge of nutrition, personal health constraints, or lack of access to dietitians.

This project proposes an **AI-Driven Diet Planning Chatbot** capable of generating personalized diet recommendations using **Natural Language Processing (NLP)** and **Machine Learning (ML)** techniques. The system interprets user inputs—such as age, weight, height, BMI, allergies, dietary preferences, lifestyle patterns, and health goals (weight loss, gain, diabetic-friendly)—and produces personalized day-wise or week-wise meal plans.

The methodology integrates **Random Forest Regression** for calorie prediction, **Decision Tree Classification** for diet type selection, **K-Means Clustering** for grouping similar profiles, and NLP models such as **BERT, spaCy, and NLTK** for contextual understanding. The chatbot communicates through a user-friendly web interface built with HTML, CSS, and JavaScript, while backend logic is handled using Django REST Framework.

This AI-powered system aims to function as a digital nutrition assistant offering instant, accurate, and personalized diet recommendations to users.

Keywords: Artificial Intelligence, Diet Planning, Machine Learning, NLP Chatbot, Calorie Prediction, Personalized Nutrition System, User Profiling.

Introduction:

1. **Background of the Problem:** Rapid urbanization, rising stress levels, and sedentary lifestyles have led to a global increase in obesity, diabetes, and nutritional deficiencies. People rarely receive personalized dietary support due to limited access to nutritionists or lack of awareness. AI-driven systems can analyze user profiles and help them adopt healthier eating habits.
2. **Importance of the Project:**
 - a. This project enables:
 1. Personalized nutrition for every individual
 2. AI-based calorie and diet prediction
 3. Contextual understanding of user needs
 4. Easy access to reliable dietary support
 5. Improved lifestyle management through guided meal planning
3. **Scope of the Project:**

The chatbot provides:

- Daily or weekly meal plans
- Calorie recommendation based on user profile

- Healthy alternatives and nutrient insights
- Support for Indian and regional diets
- Adaptation to allergies, medical conditions, and food preferences

4.Problem Statement:

Most diet applications provide generalized meal suggestions that do not cater to individual health conditions, lifestyle patterns, or cultural food habits. Rule-based chatbots fail to interpret user context, and ML models in existing apps rarely consider Indian dietary preferences.

Problem:

There is a need for an AI-powered conversational diet planning system that understands user inputs, predicts calorie requirements, recommends goal-based diets, and adapts to Indian food preferences.

Objectives

The primary objectives are:

1. Develop an AI-driven chatbot capable of personalized diet recommendation.
2. Extract user health parameters using NLP.
3. Predict daily calorie requirement using Random Forest Regressor.
4. Classify diet type using Decision Tree Classifier.
5. Cluster users using K-Means for enhanced personalization.
6. Provide daily/weekly meal plans customized to Indian food habits.
7. Build a responsive chat interface using HTML, CSS, and JavaScript.
8. Implement backend logic using Django + REST APIs.

Literature Review

1.Summary of Existing Research:

Research in the domains of Artificial Intelligence, Machine Learning, and digital Healthcare demonstrates significant progress in building intelligent dietary recommendation systems. Existing literature establishes that AI-based nutrition system can analyze large datasets, assess user characteristics, and generate meaningful dietary suggestions. Several studies highlight the use of **Machine Learning regression models**, such as Random Forest Regression, Support Vector Regression, and Gradient Boosting, for predicting calorie requirements and energy expenditure. These models typically use features such as age, gender, height, weight, BMI, and activity level to estimate the total daily calorie needs of an individual with considerable accuracy.

Similarly, classification models like **Decision Trees, Logistic Regression, K-Nearest Neighbors, and Support Vector Machines** have been employed to categorize users into various diet plans such as low-calorie, balanced, high-protein, or diabetic-friendly diets. These classification approaches help tailor nutritional recommendations according to the user's health goals. In multiple research publications, Decision Tree models have been highlighted for their interpretability, making them suitable for healthcare applications where transparency and understanding of decision-making reasoning are crucial.

Another significant contribution in existing research is the integration of **unsupervised learning algorithms**, particularly **K-Means clustering**, to group users with similar health profiles. Studies have shown that clustering helps enhance personalization because recommendations for one user in a cluster can be extended to others with similar attributes. This technique improves the accuracy and relevance of diet suggestions.

In the domain of conversational systems for healthcare, Natural Language Processing (NLP) has played a crucial role. Research on intelligent chatbots—especially healthcare assistants like Ada Health, MedWhat, and Woebot—demonstrates that conversational interfaces improve user engagement, accessibility, and compliance with health recommendations. These chatbots rely on NLP techniques such as tokenization, entity extraction, and transformer-based language models like **BERT, GPT, and spaCy pipelines**, enabling machines to understand unstructured text input from users. Transformer-based NLP models have shown excellent performance in intent classification, symptom extraction, and providing intelligent responses in healthcare settings.

Despite these advancements, most existing diet planning systems rely heavily on **Western food datasets**, limiting their applicability to the Indian population. Nutritional values, portion sizes, food preparation styles, and daily meal patterns differ significantly between regions, and existing systems do not sufficiently incorporate these cultural variations. Traditional diet recommendation apps often depend on static, rule-based engines that lack adaptability and contextual understanding.

Overall, existing research proves the reliability of AI methods—ML for prediction and classification, NLP for conversational understanding, and clustering for personalization. However, the integration of all these components into a **single, interactive, AI-driven chat-based diet planning system** remains limited.

2. Gaps Identified:

Though existing research has explored individual components of nutrition prediction and conversational health systems, several gaps persist that limit the effectiveness of AI-based dietary recommendation platforms.

a) Lack of Indian Dietary Datasets

Most AI diet applications use Western food databases, which do not account for Indian dishes like roti, dal, sabzi, idli, poha, dosa, or region-specific differences. Indian diets differ greatly in nutrients, spices, and preparation methods. Therefore, existing models fail to deliver culturally accurate recommendations for Indian users.

b) Limited Integration of NLP and ML

Many systems use **NLP-only rule-based chatbots**, which cannot understand complex user statements. Conversely, systems using ML for prediction often lack conversational capabilities. Very few studies integrate NLP (intent recognition + parameter extraction) with ML (calorie prediction + diet classification) into a unified end-to-end system.

c) Inability to Extract Detailed User Information

Existing chatbots struggle to extract structured health parameters like age, weight, height, diet preference, allergies, and medical conditions from natural sentences such as: *"I am 21 years old, 68kg, vegetarian, and trying to lose weight."* Advanced NLP techniques are required, but most traditional systems do not employ them.

d) Lack of Personalization and Adaptiveness

Most diet applications provide **generic meal plans**, failing to consider:

- user's lifestyle
- allergy constraints
- medical conditions (diabetes, thyroid, PCOS)
- daily activity level
- food preferences (veg, non-veg, vegan)

This reduces the usefulness of such applications in real-world health management.

e) Minimal Use of Hybrid AI Architectures

There is limited research on combining:

- **Regression models** for calorie prediction
- **Classification models** for diet type selection
- **Clustering models** for grouping similar users
- **NLP-based chat interfaces** for conversation

Most existing systems rely on only one or two components rather than integrating all into a full-stack diet recommendation system.

f) Poor User Engagement in Existing Systems

Research indicates that users respond better to interactive systems that can converse naturally rather than apps that require manual input. However, existing diet planning platforms lack conversational interfaces capable of guiding users step-by-step, reducing long-term engagement.

g) Limited Support for Real-Time Query Handling

Current systems often provide static diet charts rather than dynamic, real-time diet suggestions based on live user conversations.

Methodology

Detailed Explanation:

The methodology for developing the AI-Driven Diet Planning Chatbot is divided into eight key stages:

1. Data Preparation:

User profile data (age, weight, height, gender, preferences) and Indian food nutrition datasets are collected. The data is cleaned by removing duplicates, handling missing values, standardizing units, and encoding categorical variables. The final dataset is split into training and testing sets.

2. Exploratory Data Analysis (EDA):

Statistical charts and visualizations are used to study BMI patterns, calorie distributions, nutrient categories, and correlations between variables like age, weight, and caloric needs. Outliers and inconsistencies are identified and corrected.

3. Feature Engineering:

New features such as BMI, BMR, and TDEE are computed. Food-related features like calorie density and nutrient ratio are engineered. Text inputs from users are preprocessed using tokenization, lemmatization, and stop-word removal for NLP tasks.

4. Survival Analysis Modeling:

Survival analysis is adapted to estimate how long users typically follow a diet plan. Kaplan–Meier curves and Cox models help identify factors affecting diet adherence, such as calorie deficit, age, and food preference.

5. Deep Learning Modeling:

A transformer-based NLP model (BERT) is used to understand user messages and extract age, weight, goals, allergies, and dietary preferences. A neural network is used to score and recommend meal options based on user constraints.

6. Hybrid Model Combination:

Machine Learning (Random Forest, Decision Tree, K-Means), Deep Learning (BERT), and rule-based logic are combined. ML predicts calorie needs, NLP extracts user details, and a rule engine generates personalized Indian meals.

7. Model Evaluation:

Models are evaluated using MAE, MSE, accuracy, F1-score, silhouette score, and NLP F1 metrics. The hybrid model shows improved prediction accuracy and conversation quality.

8. Model Interpretation:

Feature importance, SHAP values, decision paths, PCA visualization, and BERT attention heatmaps help interpret model decisions. This makes the recommendations transparent and reliable.

Tools, Technologies, and Resources:

The development of the AI-Driven Diet Planning Chatbot requires a combination of software tools, programming libraries, and data resources to successfully implement machine learning models, NLP pipelines, and a functional web-based chatbot interface.

1. Programming Languages

- **Python** – used for Machine Learning, NLP, model training, and backend logic.
- **JavaScript** – used to handle chatbot interactions on the frontend.
- **HTML & CSS** – used to design the chat interface and layout.

2. Frameworks & Libraries

Machine Learning & Data Processing

- **Scikit-learn** – for Random Forest, Decision Tree, K-Means, and preprocessing.
- **Pandas, NumPy** – for data manipulation and cleaning.
- **Matplotlib, Seaborn** – for visualization and EDA.

Deep Learning & NLP

- **TensorFlow / PyTorch** – for neural network-based recommendation layers.
- **BERT / HuggingFace Transformers** – for intent detection and entity extraction.
- **spaCy, NLTK** – for tokenization, lemmatization, and text preprocessing.

Web Development

- **Django** – backend framework used to build APIs and manage user requests.
- **Django REST Framework** – for creating RESTful endpoints for chatbot communication.

3. Tools

- **VS Code / PyCharm** – development environment for coding.
- **Jupyter Notebook** – used for experiments, EDA, and ML model testing.
- **Postman** – API testing and validation.
- **Git & GitHub** – version control and project collaboration.

4. Databases

- **SQLite / MySQL** – to store user profiles, diet plans, and model outputs.

5 . Datasets & Resources

- Indian Food Calorie Dataset
- User Profile & Goal Dataset
- Nutritional Information Databases
- NLP training corpus for diet-related text

6. Hardware Requirements

- A system with at least:
 - 8 GB RAM
 - Multi-core CPU
 - Optional: GPU for faster NLP model training

7. Additional Resources

- Research papers, nutritional references, and food science documents.
- Open-source repositories for diet datasets and model benchmarking.

❖ **Workflow or Project Plan:**

The project follows a structured workflow beginning with requirement analysis, where user needs and system objectives are identified. Data is collected from Indian food nutrition datasets and user health profiles, followed by data preparation that includes cleaning, encoding, and standardization. Exploratory Data Analysis (EDA) is performed to understand patterns in BMI, calorie distribution, and user behavior trends.

Feature engineering is then applied to compute BMI, BMR, TDEE, nutrient ratios, and to preprocess text for NLP tasks. Machine Learning models such as Random Forest, Decision Tree, and K-Means are developed for calorie prediction, diet classification, and user

clustering. In parallel, NLP models like BERT and spaCy are used to extract user intent and health parameters from natural language input.

A hybrid system integrates ML predictions, NLP outputs, and a rule-based diet generator to provide personalized diet plans. Backend APIs are built using Django REST Framework and connected to a responsive chat-based frontend developed with HTML, CSS, and JavaScript. The system undergoes model evaluation, testing, and refinement before final deployment and documentation.

Implementation Plan

1. **Requirement Analysis:** Identify user needs, system objectives, and project components for AI-based diet planning.
2. **Dataset Collection & Preprocessing:** Gather nutrition and user datasets, clean data, standardize units, and encode variables.
3. **Model Training for Intent Classification:** Train NLP model (BERT/spaCy) to detect user intent and extract diet-related inputs.
4. **Development of Diet Recommendation Module:** Build ML models and rule-based logic to generate personalized diet plans.
5. **Chatbot Response Logic & Integration:** Integrate NLP, ML outputs, and conversational flows into backend chatbot logic.
6. **Frontend UI Creation:** Develop chat interface using HTML/CSS/JS and connect it to backend APIs.
7. **Backend API and Deployment:** Implement Django REST APIs, load trained models, and deploy the complete system.
8. **Evaluation and Final Presentation:** Test model accuracy, refine responses, prepare documentation, and present the final system

Expected Outcomes

- **Improved Diet Recommendation Accuracy:** The system is expected to deliver highly accurate and personalized diet plans by combining ML-based calorie prediction, NLP-based intent understanding, and rule-based meal mapping.
- **Deeper Health & Nutrition Insights:** Users will receive meaningful insights into their BMI, calorie needs, nutrient intake, and dietary patterns, helping them understand their overall health better.
- **Increased Nutrition Awareness Levels:** Through interactive chatbot conversations, users will gain awareness about healthy food choices, balanced diets, low-calorie alternatives, and

lifestyle-specific dietary habits.

Project Timeline (20 Nov 2025 – 17 Dec 2025)

Week / Dates	Activities Performed
Week1 20 Nov – 23 Nov 2025	Requirement gathering, Project approval & guidance discussion
Week2 24 Nov – 27 Nov 2025	Data collection, Indian food nutrition dataset preparation
Week3 28 Nov – 30 Nov 2025	Data preprocessing, Feature engineering, BMI & calorie formulas setup
Week4 1 Dec – 5 Dec 2025	ML Model development: Calorie prediction & Diet classification
Week5 6 Dec – 9 Dec 2025	NLP chatbot interaction design and integration
Week6 10 Dec – 13 Dec 2025	Frontend chat interface + Django REST API integration
Week7 14 Dec – 16 Dec 2025	Testing, refinement, performance optimization
Final Day 17 Dec 2025	Final Deployment, Documentation, Synopsis submission